MPG No. 7.1.20 $mm2$ [Type I, orthorhombic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{M}^{(1)}_{xy}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}^{(1)}_{z}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{yzx} \end{bmatrix}$$

$$M_{yzx} = \mathbb{M}^{(1)}_{xyz}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & M_{zxy} & 0 \\ 0 & 0 & -M_{yzx} - M_{zxy} \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}^{(1)}_{u} + \mathbb{T}^{(1)}_{v}$$
$$M_{zxy} = -3\mathbb{T}^{(1)}_{u} + \mathbb{T}^{(1)}_{v}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{zxy} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(2)}_{u} + \mathbb{T}^{(2)}_{v}$$
$$M_{zxy} = \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(2)}_{u} - \mathbb{T}^{(2)}_{v}$$
$$M_{xyz} = \mathbb{T}^{(1)}_{0} + 2\mathbb{T}^{(2)}_{u}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xxxy} & M_{yyxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ -M_{xxxy} & -M_{yyxy} & -M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_{z}^{(1)} - 2\mathbb{T}_{z}^{\alpha(1)} + 2\mathbb{T}_{z}^{\beta(1)}$$
$$M_{yyxy} = \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_{z}^{(1)} + 2\mathbb{T}_{z}^{\alpha(1)} + 2\mathbb{T}_{z}^{\beta(1)}$$
$$M_{zzxy} = \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_{z}^{\beta(1)}$$
$$M_{yzzx} = -\mathbb{T}_{z}^{(1)} - 4\mathbb{T}_{z}^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_{z}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & M_{yyxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yzzx} & 0 \\ M_{zxyz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{(3)} - \mathbb{T}_{z}^{\alpha(2)} + \mathbb{T}_{z}^{\beta(2)}$$
$$M_{yyxy} = 2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{(3)} - \mathbb{T}_{z}^{\alpha(2)} - \mathbb{T}_{z}^{\beta(2)}$$
$$M_{zzxy} = \mathbb{T}_{z}^{(3)} + 2\mathbb{T}_{z}^{(4)} + 2\mathbb{T}_{z}^{\alpha(2)}$$
$$M_{yzzx} = -\mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{(4)} - \mathbb{T}_{z}^{\alpha(2)} - \mathbb{T}_{z}^{\beta(2)}$$
$$M_{zxyz} = \mathbb{M}_{xy}^{(5)} + \mathbb{T}_{z}^{(4)} - \mathbb{T}_{z}^{\alpha(2)} + \mathbb{T}_{z}^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & 0 \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)}$$
$$M_{zxyz} = -\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)}$$
$$M_{xyxx} = -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)}$$
$$M_{xyyy} = 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)}$$
$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{yyxy} & 0 & 0 & M_{yyyx} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzyx} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & 0 & M_{zxzy} & 0 & 0 \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$
$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} + 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{yyxy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{yyyx} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)}$$
$$M_{zzyx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{(4)} - 2\mathbb{T}_z^{\alpha(2)} - 4\mathbb{T}_z^{\beta(1)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} - \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{yzxz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} - \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$
$$M_{zxyz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} + \mathbb{T}_z^{\beta(2)}$$
$$M_{zxzy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} + 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} - \mathbb{T}_z^{\beta(2)}$$
$$M_{xyxx} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - 2\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)} - 2\mathbb{T}_z^{\beta(1)}$$
$$M_{xyyy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} - 2\mathbb{T}_z^{\beta(1)}$$
$$M_{xyzz} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} - \mathbb{M}_{xy}^{(3)} + 4\mathbb{T}_z^{\beta(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & 0 & M_{zxzy} & 0 & 0 \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)}$$
$$M_{yzxz} = -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)}$$
$$M_{zxyz} = 2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)}$$
$$M_{zxzy} = -2\mathbb{M}_{xy}^{(4)} - \mathbb{M}_{xy}^{(6)} - \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)}$$
$$M_{xyxx} = -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} + \mathbb{T}_z^{\beta(3)}$$
$$M_{xyyy} = 2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} - \mathbb{T}_z^{\beta(3)}$$
$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & M_{xxxy} & 0 \\ M_{yyyx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzxy} & 0 \\ M_{xxyx} & 0 & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{xyyy} & 0 \\ 0 & 0 & \frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} - M_{yyyx} + M_{zzxy} \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)}$$
$$M_{yyyx} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)}$$
$$M_{zzxy} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)}$$
$$M_{xxyx} = \mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)}$$
$$M_{yzzx} = 2\mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)}$$
$$M_{xyyy} = -\mathbb{M}_{4z}^{\alpha(1)} - \mathbb{M}_{4z}^{\beta(1)} + \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)}$$